

PlanComplexity / PyUCoMX

Current application summary - research use only

Purpose and workflow

Local RTPLAN complexity analysis for medical-physics research. The Tk desktop GUI supports single files, folder batches, AUTO or explicit modes, metric notes, warnings and CSV export. READY means the analysis path is supported; individual metrics may still be unavailable.

Mode	Catalog keys	Current formula version
VMAT_IMRT	286	geometry-v4
TOMO	65	tomo-v3
CYBERKNIFE_MLC	6	geometry-v4
AURORA	77	aurora-v4-physical-aperture-open-only

Metric boundaries

Catalog counts include layer variants and container keys; actual runtime outputs depend on geometry. Halcyon/Ethos native, effective and nonphysical stacked representations have distinct definitions. CyberKnife is a six-metric MLC subset. TOMO motion is planned, not measured delivery.

Aurora: 77 keys and open-only physical aperture metrics

The existing 70 research/proxy keys are retained, plus seven V4 descriptors: mean_ba, mean_bi, mean_ca, mean_sas5, mean_sas10, mean_mcs_aurora and mcs_complexity_aurora. V4 uses real leaf boundaries, A/B banks, dual-layer intersection and X/Y jaw clipping. All seven means exclude area-zero endpoints and renormalize retained MU weights. SAS/LSV exclude closed gaps. Exclusion count/weight fraction and missing-MU fallback are disclosed in warnings. No open endpoint makes all seven unavailable. Endpoint sampling is not continuous delivery reconstruction. The adapted MCS is not classic McNiven MCS; historical V4 versions must not be mixed.

Run, export and package

Install: `python -m pip install -r requirements.txt`. GUI: `python ucomx.py`. Single-plan console: `python main.py --input-file path/to/plan.dcm`. Main GUI/service export writes `results.csv` and `results_columns.csv`. The standalone `aurora_svmat_cli.py` can export separate plan, beam and trajectory tables. Windows packaging uses `tools/build_windows_exe.ps1` and `PyUCoMX.spec`, producing `dist/PyUCoMX.exe`. Rebuilding does not publish a GitHub release.

Reproducibility and documentation

Keep input identity/hash, source commit, mode, formula version and precision with study results. Use `full_precision=True` for VMAT/IMRT and CyberKnife validation. CSV/DICOM clinical data remain local. Current instructions: `docs/user_guide.md` and `docs/README.md`. Full formulas: `docs/metric_formula_contracts.md`; Aurora details: `docs/aurora_aperture_metrics.md`. Regression results are technical evidence, not clinical validation.